

Ethanol Blending in Gasoline – Netherlands

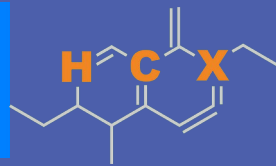
July 2026



**U.S. GRAINS &
BIOPRODUCTS
COUNCIL**

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90

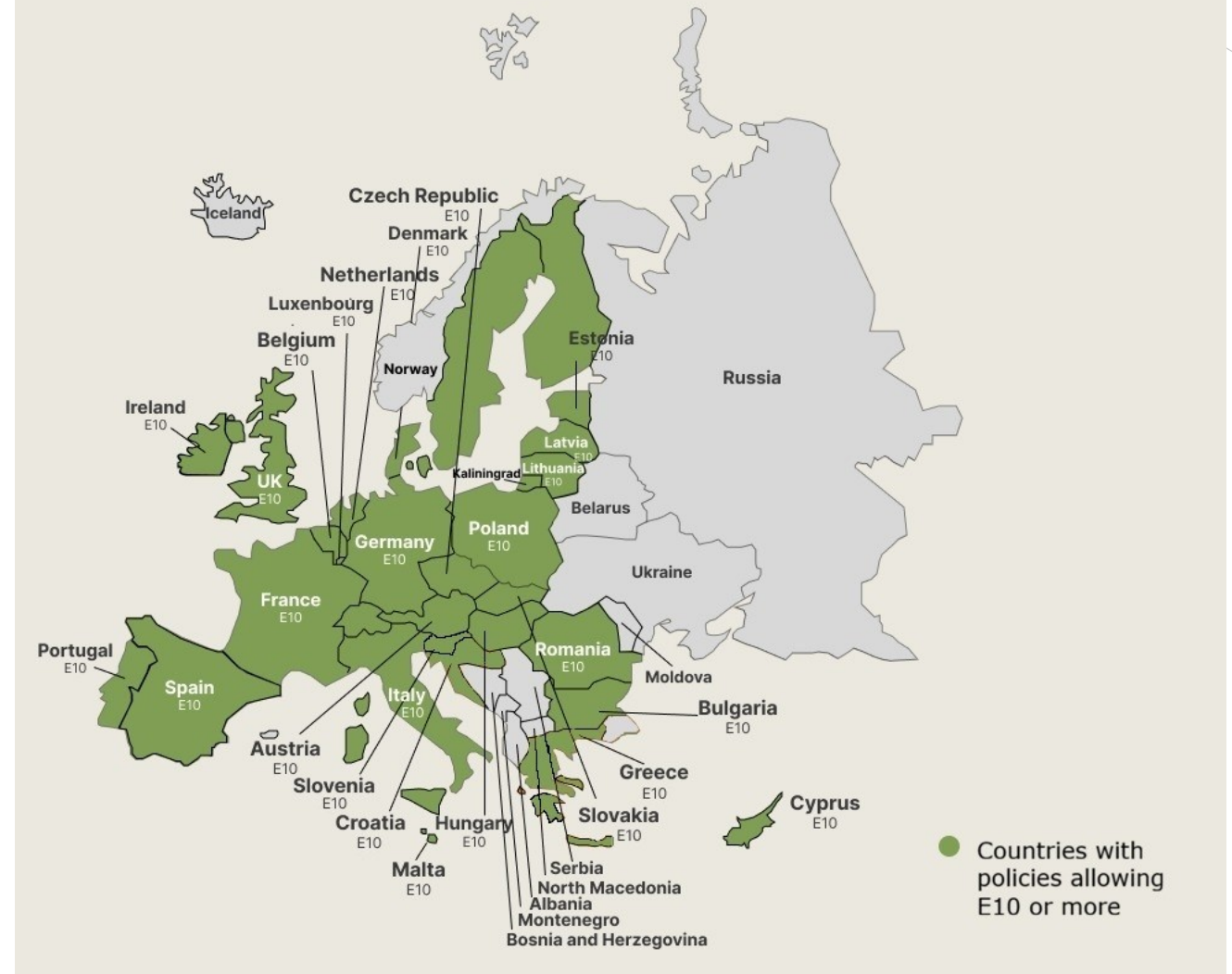


www.faro90.com
+52 55 6122 2255

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

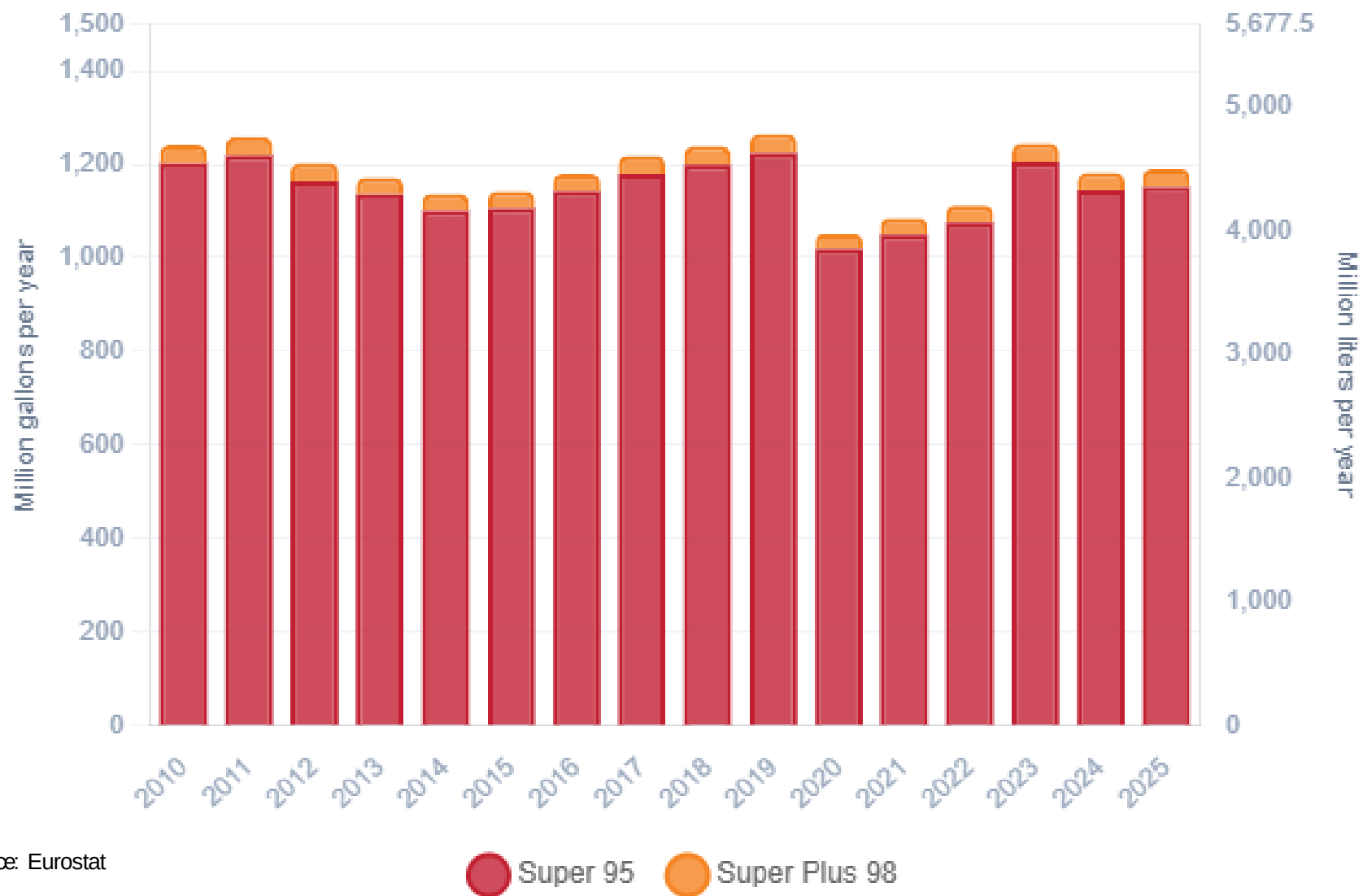
Ethanol Gasoline Blending in Netherlands



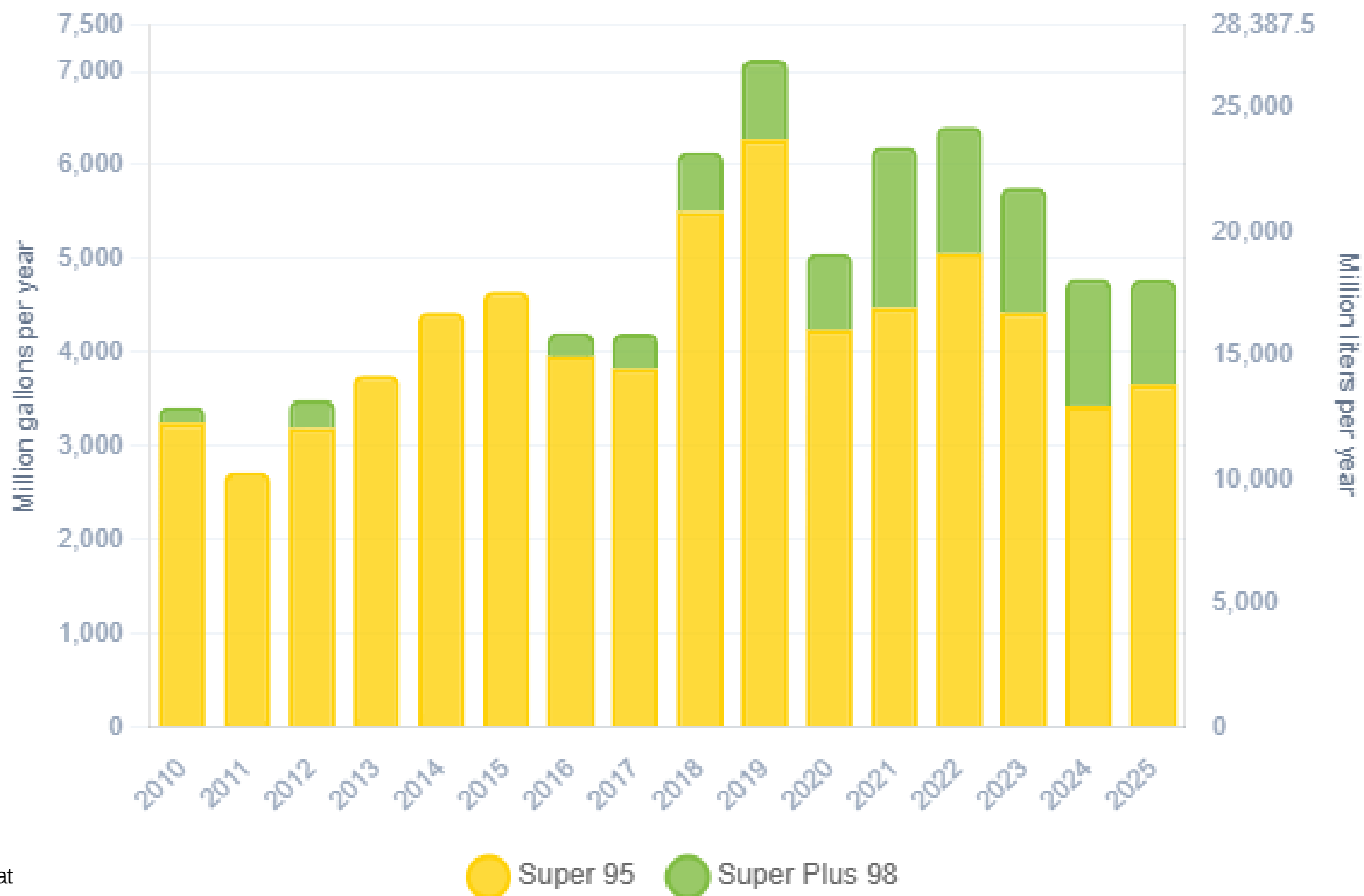
In 2025, gasoline consumption in Netherlands was 4,514 million liters (1,192 million gallons) and growth rate was -0.28%. Gasoline production and net imports were 18,087 and -13,573 million liters (4,778 and -3,586 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.1 RON.

In 2025, ethanol consumption in Netherlands was 500 million liters (132 million gallons) representing 11% of gasoline demand. Ethanol Net Imports were 695 million liters (0 and 184 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

Gasoline Demand in Netherlands



Gasoline Production in Netherlands



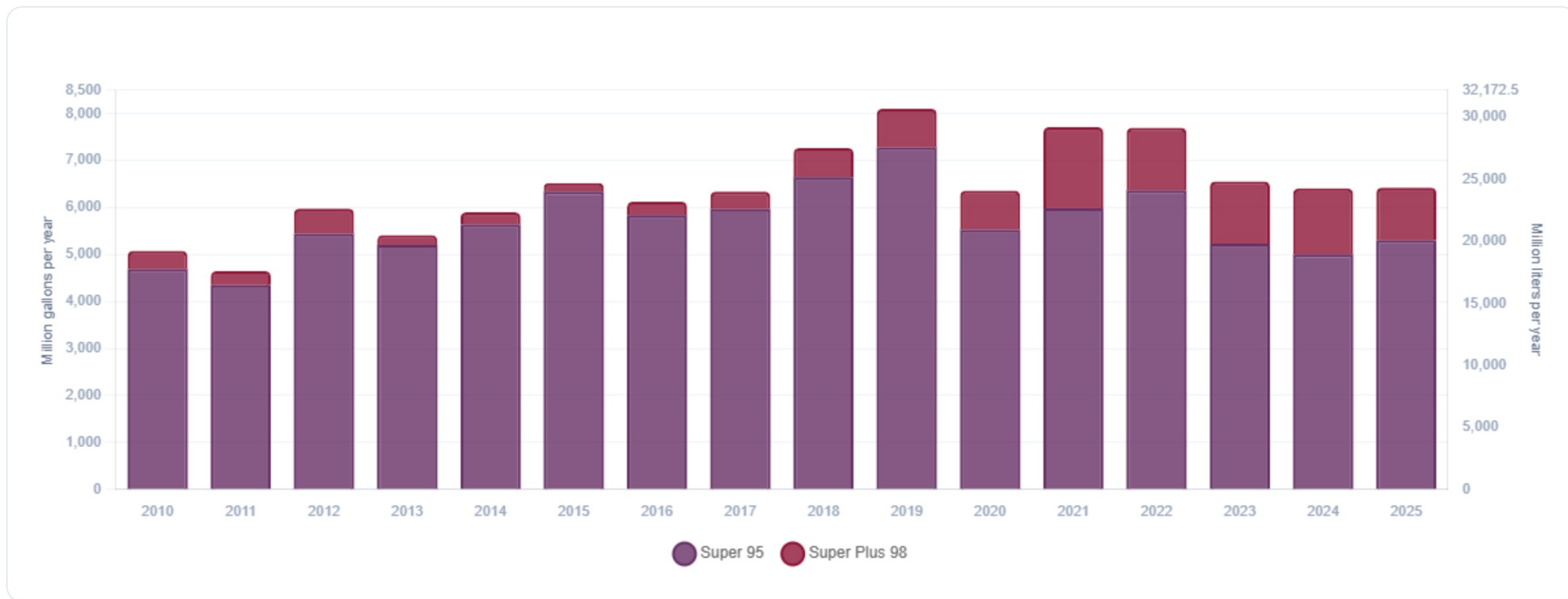
Source: Eurostat

Gasoline Imports in Netherlands



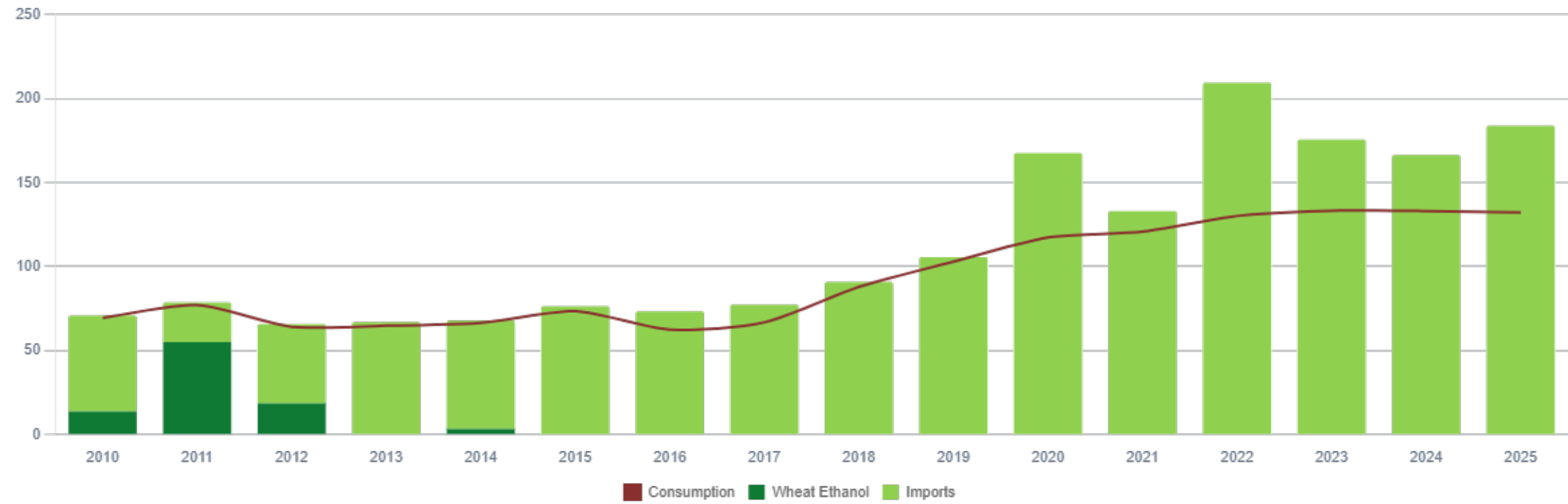
Source: Eurostat

Gasoline Exports in Netherlands



Source: Eurostat

Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

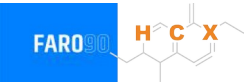
Gasoline Quality in Netherlands

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

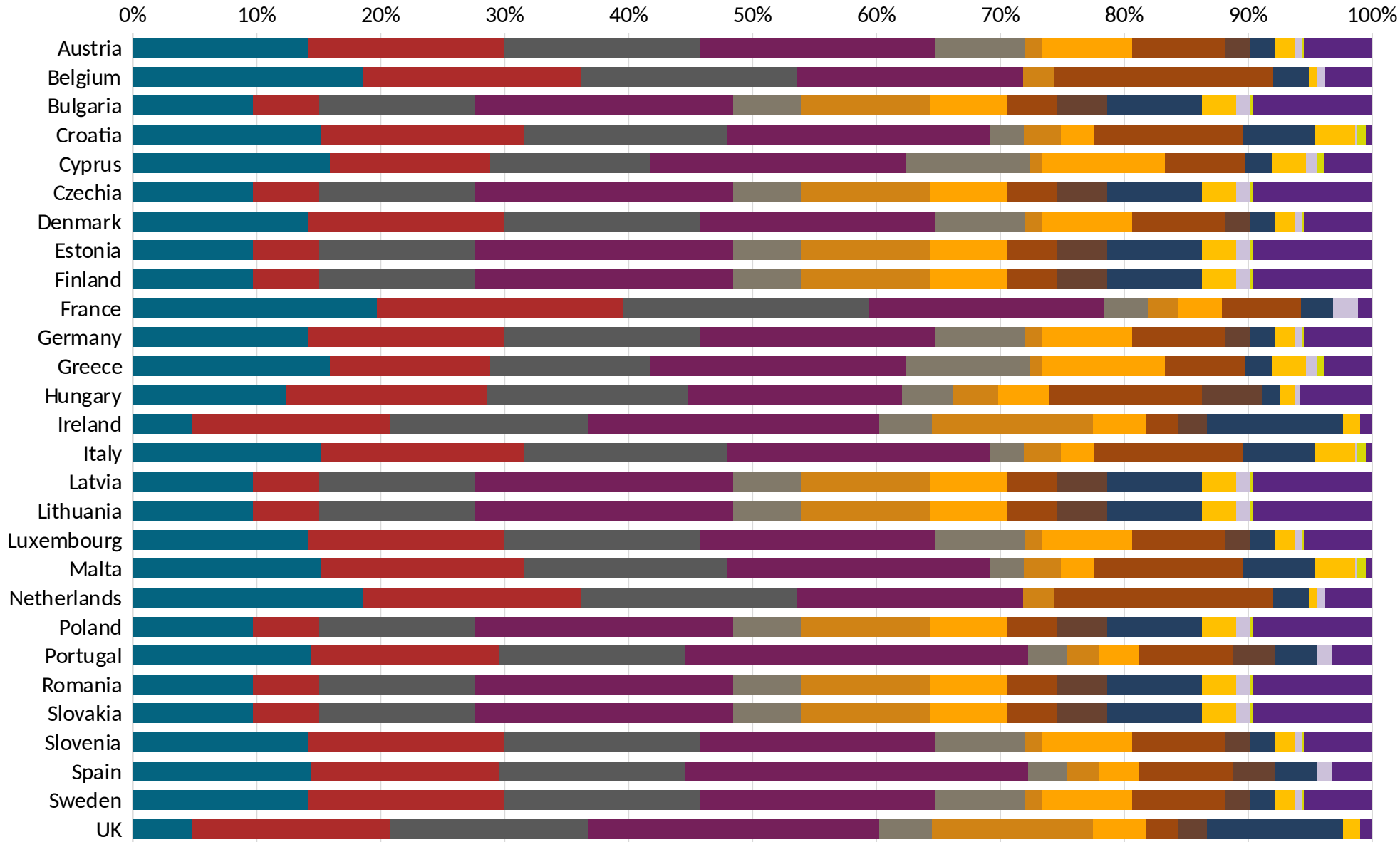
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



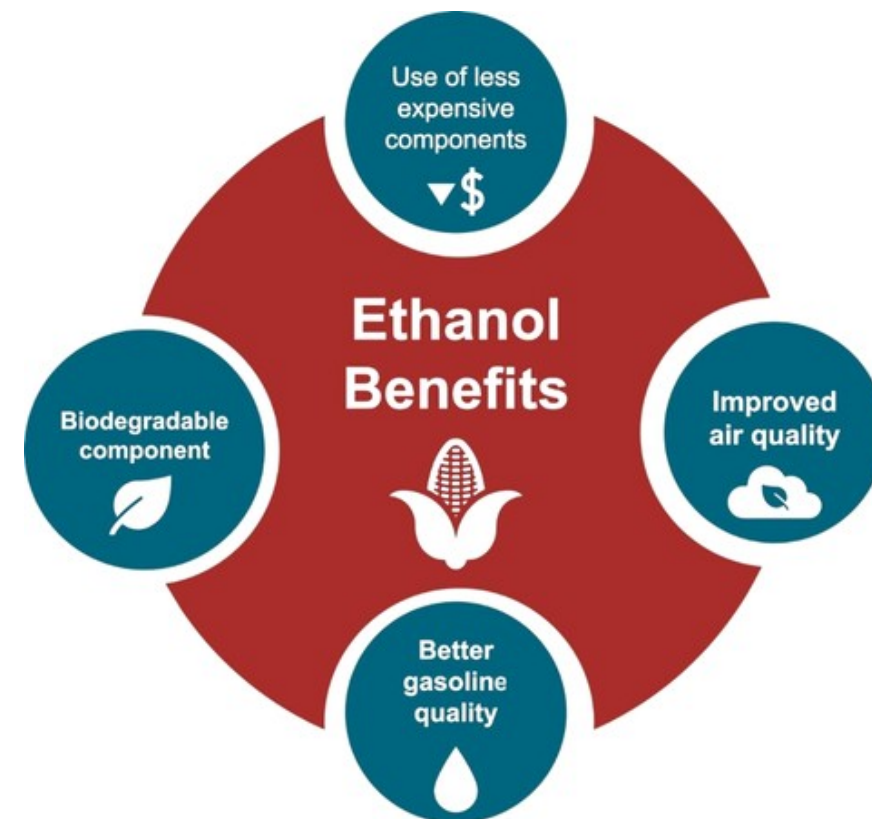
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

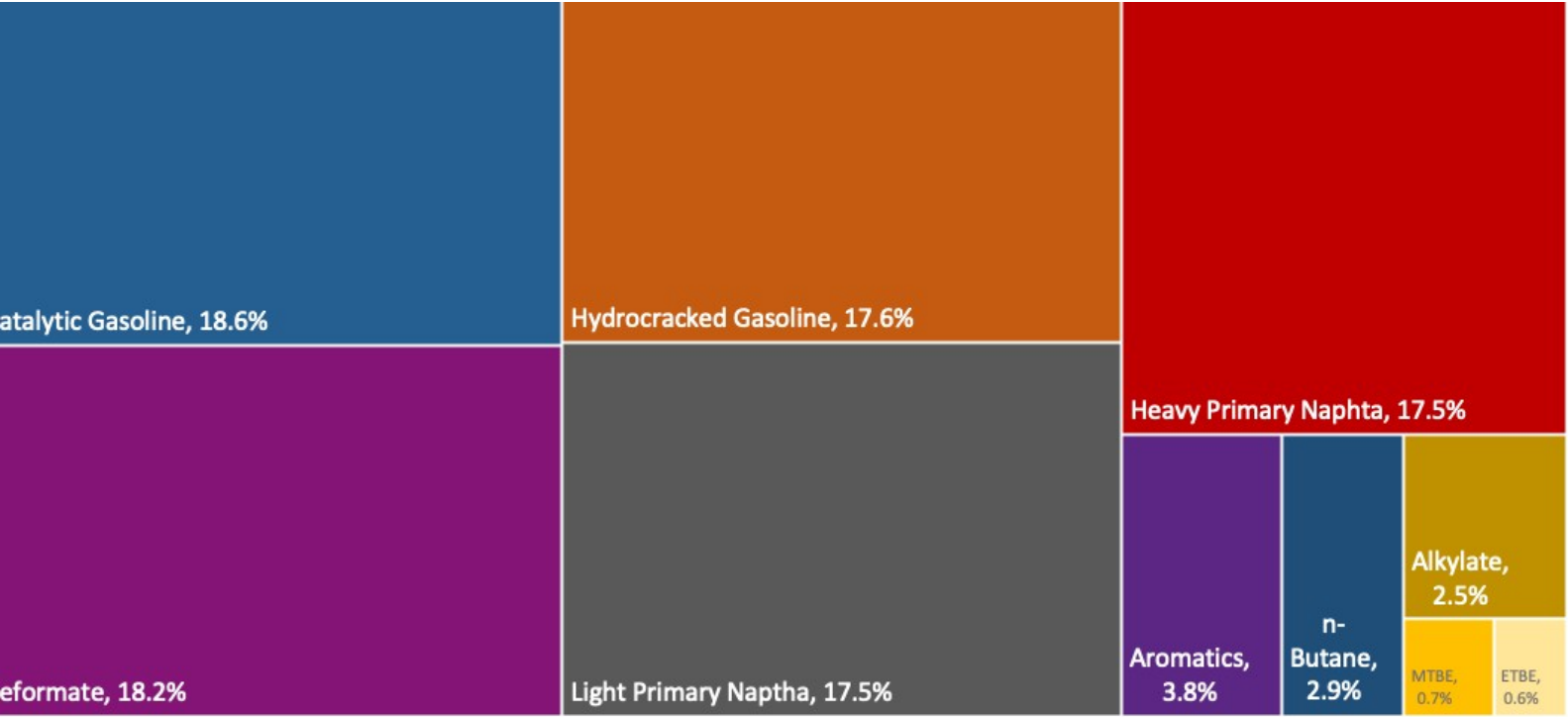
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

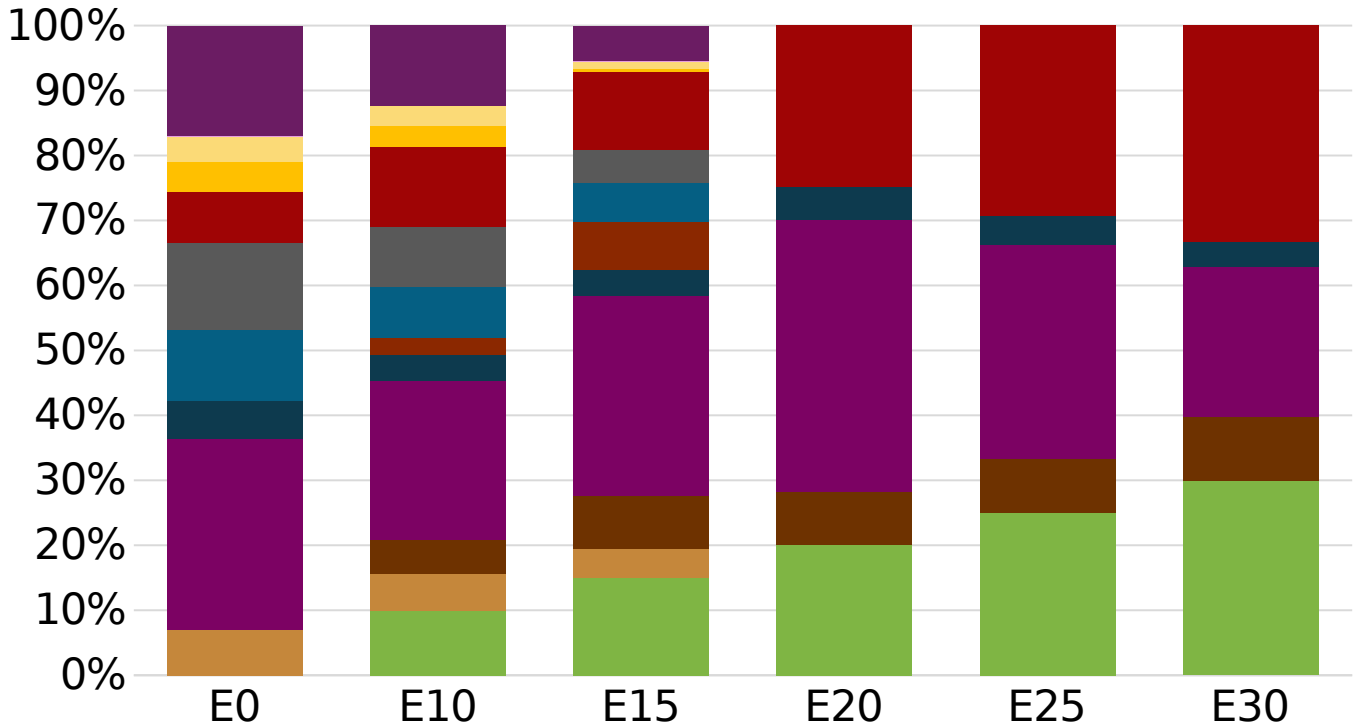
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Netherlands Available Blending Components



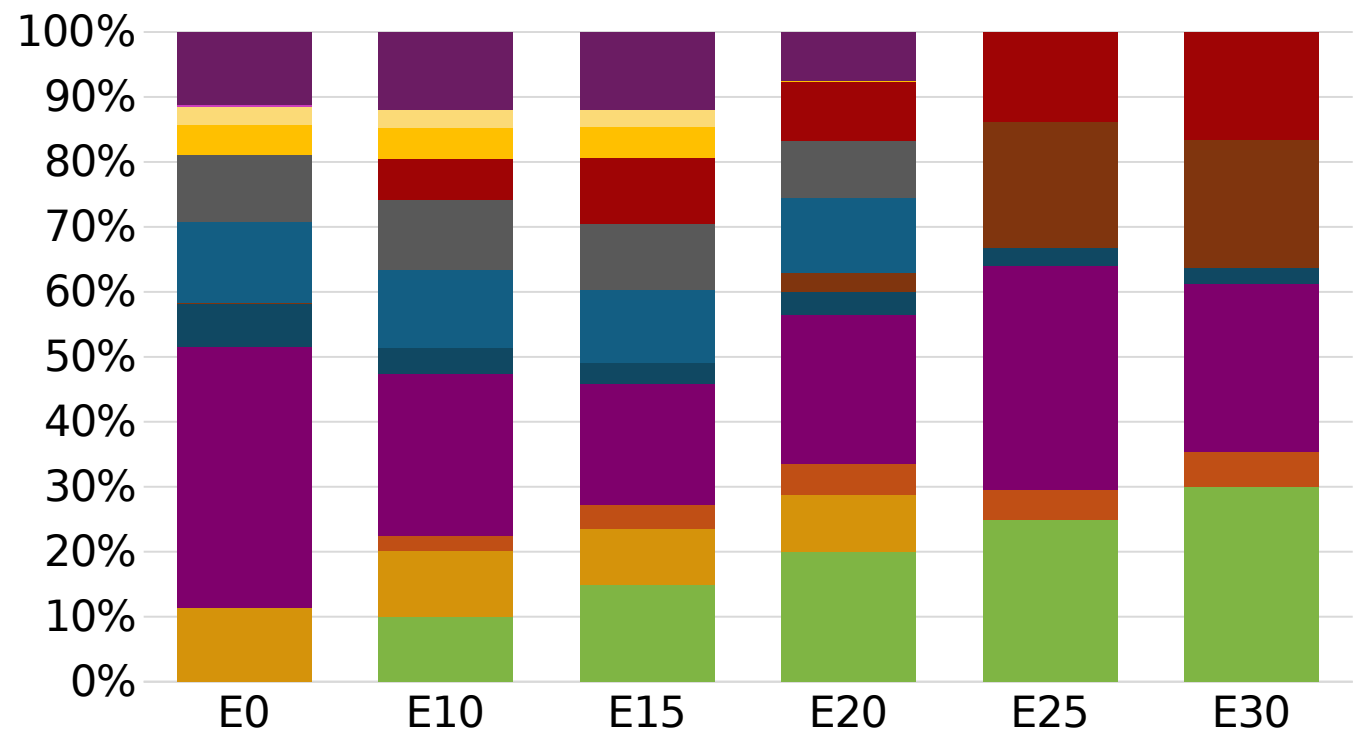
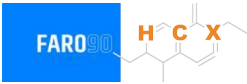
Netherlands – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.078	\$ 1.944	\$ 1.869	\$ 1.748	\$ 1.672	\$ 1.577

Netherlands - Gasoline 98 - Constant Octane

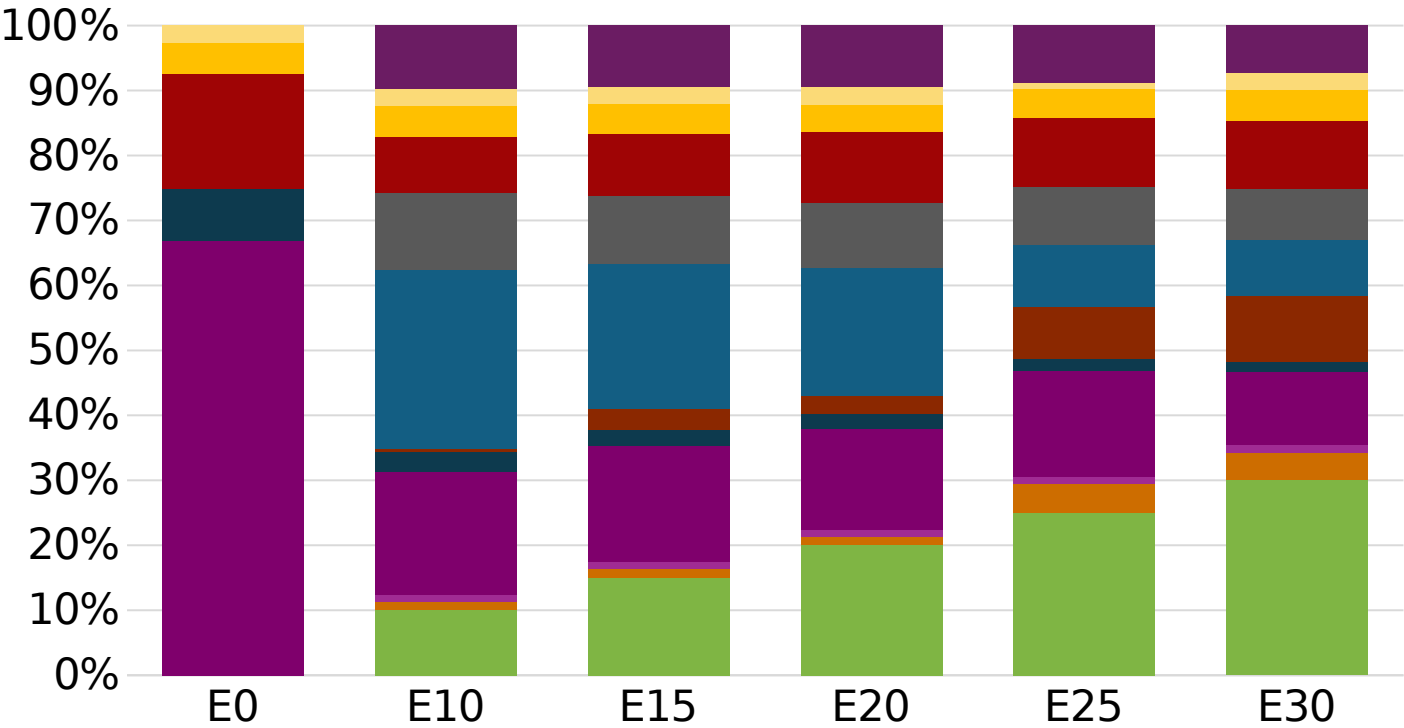


- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.206	\$ 2.087	\$ 2.013	\$ 1.961	\$ 1.854	\$ 1.791

Average CIF 2025 Prices

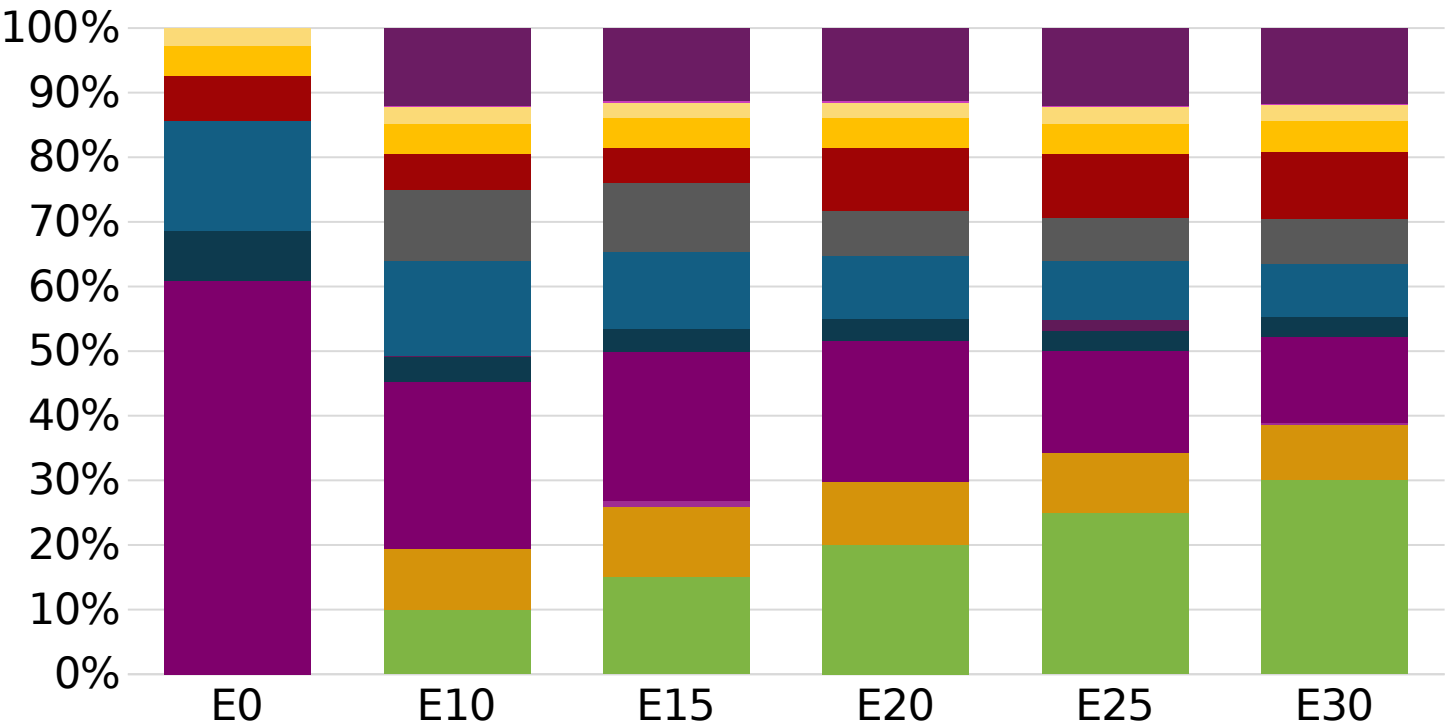
Netherlands – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990

Netherlands – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

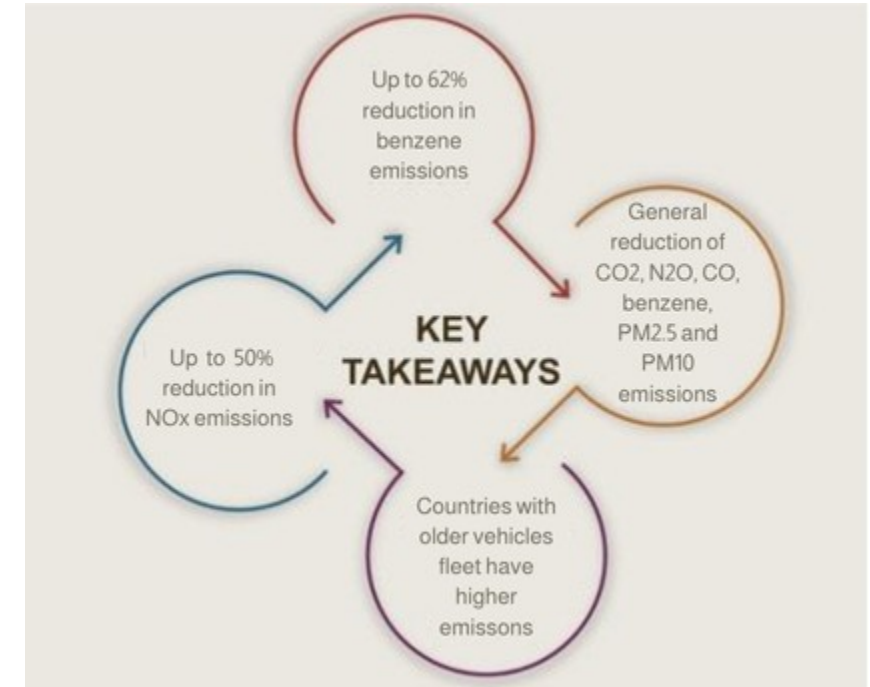
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

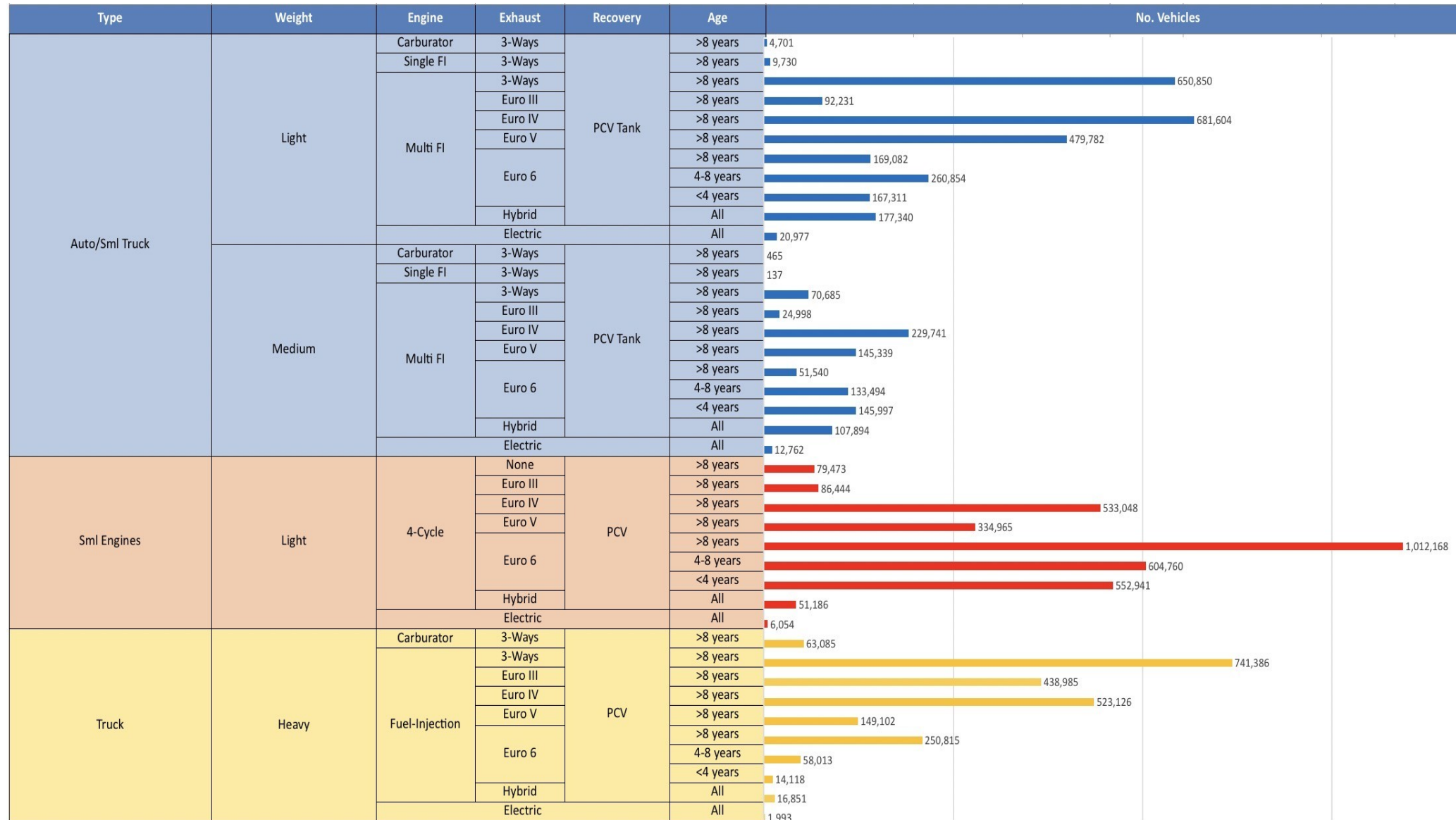
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Netherlands

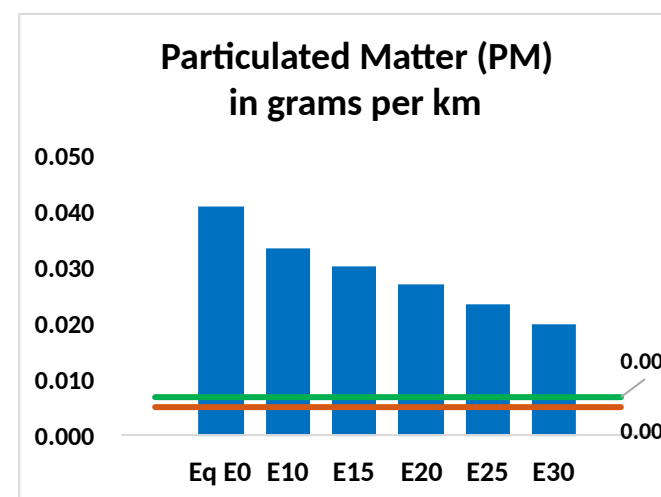
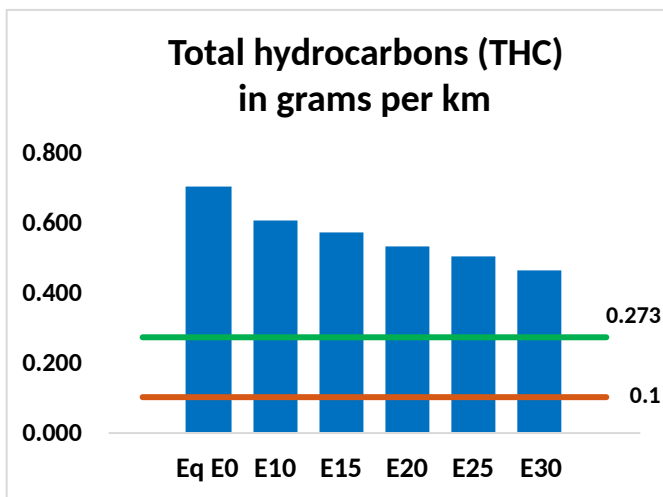
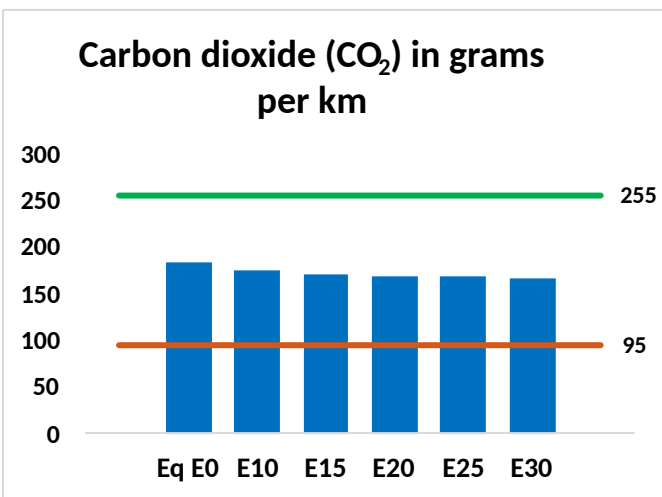
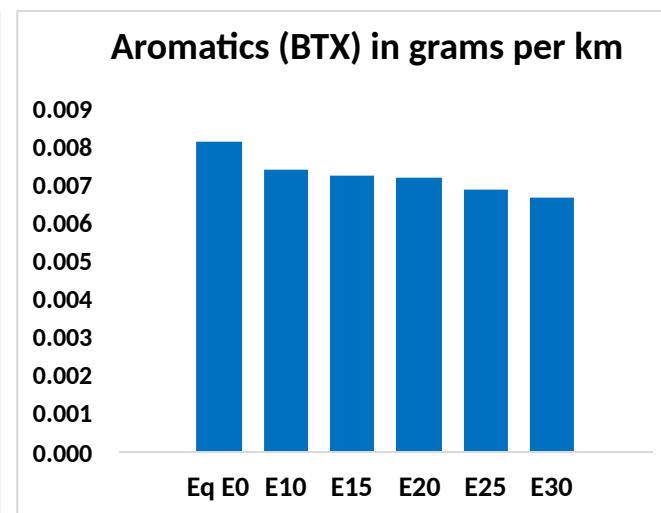
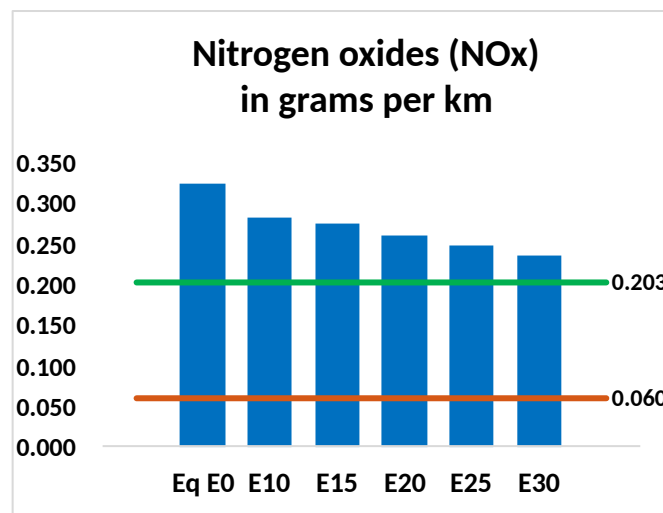
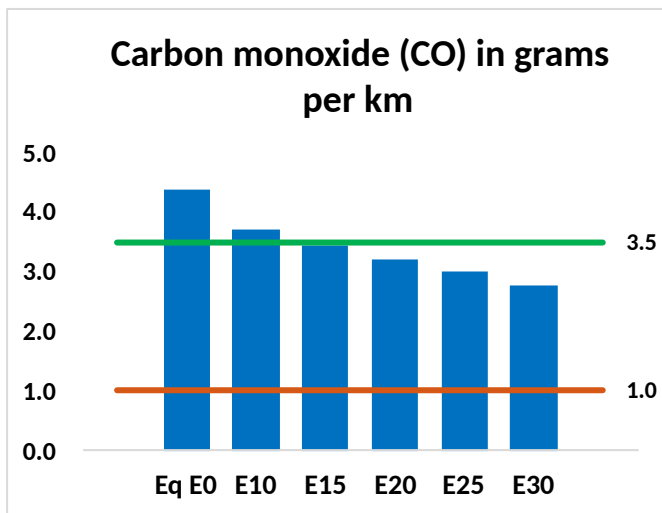
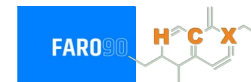


Vehicle Fleet: 9,156,027
Gasoline Vehicle Fleet: 7,160,013

Source: Eurostat, ACEA

Netherlands – Vehicular Emissions

— TIER III BIN 125 United States
— Euro 6 Standard



Netherlands – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	641,117	593,663	546,209	507,680	469,072	439,434	406,683	-15%	-27%
CO2	27,000,283	26,305,990	25,650,269	25,134,563	24,879,372	24,767,928	24,311,384	-5%	-8%
VOC	103,136	96,195	89,254	83,759	78,317	74,160	68,518	-13%	-24%
NOx	47,938	44,263	41,727	40,428	38,343	36,705	34,778	-13%	-20%
SOx	306	283	260	240	222	201	180	-15%	-28%
NH3	10,439	10,335	10,232	10,280	10,396	10,477	10,545	-2%	0%
N2O	450	447	445	448	457	462	467	-1%	2%
CH4	22,880	22,880	22,880	23,223	23,727	24,161	24,573	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	671	645	619	603	589	579	563	-8%	-12%
Acetaldehyde	1,960	2,249	3,300	5,025	6,846	7,823	9,244	68%	249%
Formaldehyde	7,275	7,073	8,245	9,682	10,143	11,221	12,215	13%	39%
Benzene	1,193	1,146	1,086	1,069	1,060	1,014	981	-9%	-11%
PM2.5	1,667	1,515	1,364	1,230	1,100	960	814	-18%	-34%
PM10	4,343	3,948	3,553	3,206	2,866	2,502	2,120	-18%	-34%

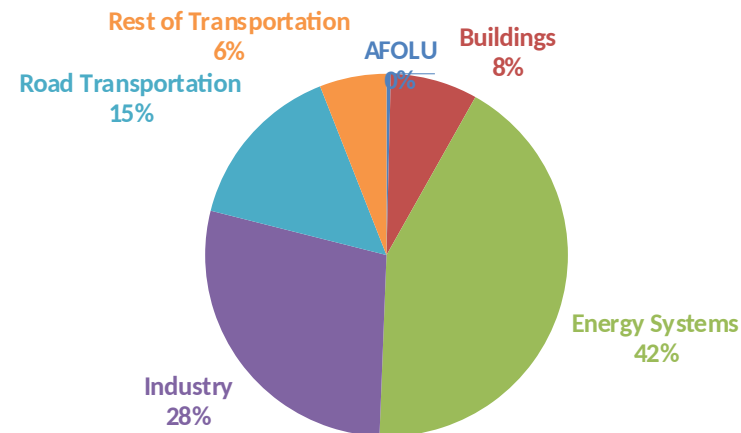
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

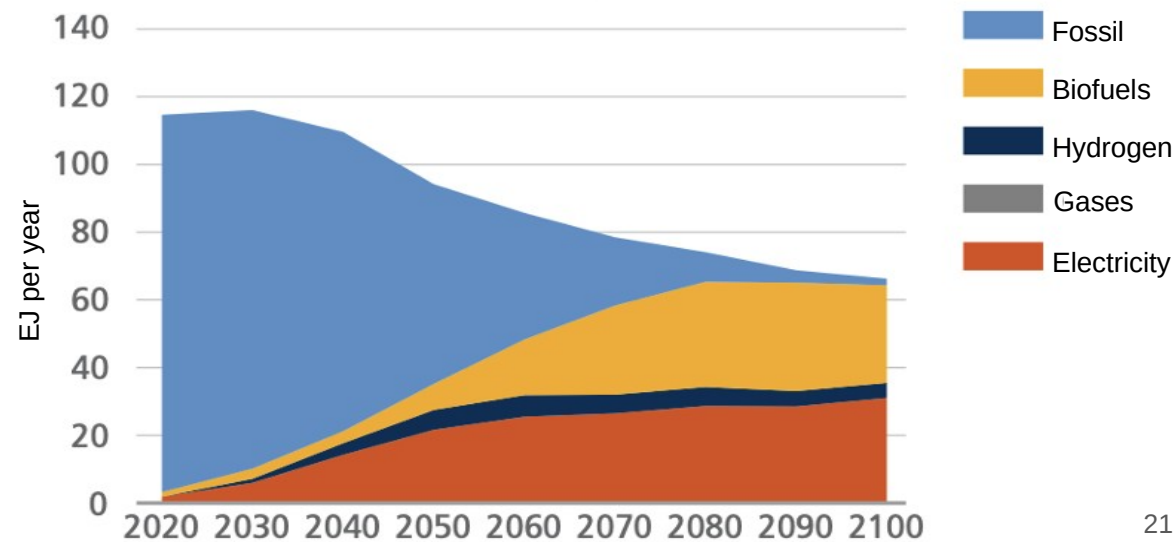
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

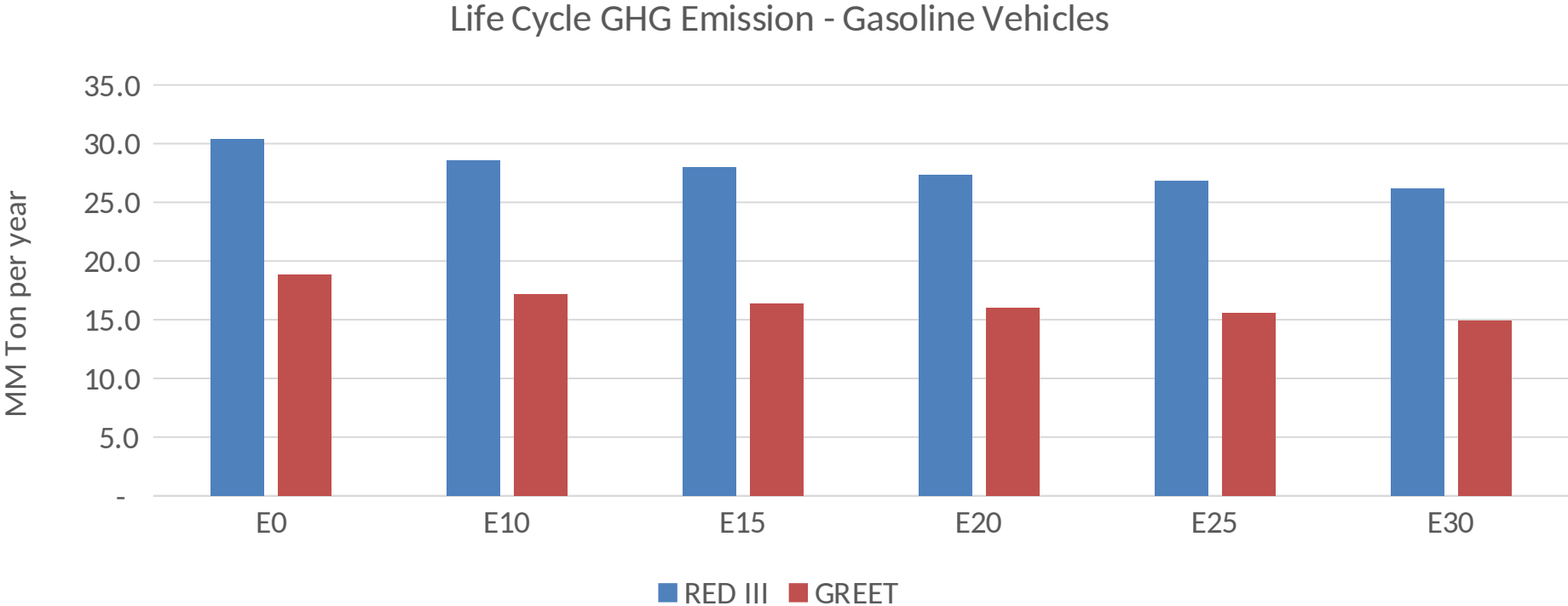
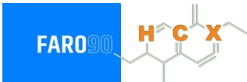
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.8%	13.0%	14.8%	17.3%	20.7%
RED II/III	0.0%	6.0%	8.0%	10.1%	11.8%	14.0%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.2%	3.3%	3.7%	4.3%	5.2%
RED II/III	0.0%	2.4%	3.3%	4.1%	4.8%	5.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90